

Drag to pan · click a part to trace its wiring · Esc to clear



Wire	From	PIN	TO	PIN	FUNCTION
	Ryobi 18V Battery	Positive (+)	30A Inline Blade Fuse	Input	Thick Power Line
	30A Inline Fuse	Output	Main Power Switch	Input	Thick Power Line
	Main Power Switch	Output	Primary 2-in-10 Lever	Positive Input (1 in)	Main 18V Power Feed
	Ryobi 18V Battery	Negative (-)	Primary 2-in-10 Lever	Negative Input (1 in)	Main 18V Ground Feed
	Primary 18V Hub	Output Load 1	Left Leg Flipsky ESC	Power Input (V+/GND)	18V High-Current Drive
	Primary 18V Hub	Output Load 2	Right Leg Flipsky ESC	Power Input (V+/GND)	18V High-Current Drive
	Primary 18V Hub	Output Load 3	12V/8A Buck Converter	Input (V+/GND)	18V Sub-System Supply
	Primary 18V Hub	Output Load 4	5V Buck Converter	Input (V+/GND)	18V Logic Supply
	Primary 18V Hub	Output Load 5	Slip Ring Crossover	Chassis Side Base	18V Dome Main Line
	12V/8A Buck Converter	Output	Secondary 12V Hub	Input	Inferred: not in list
	Secondary 12V Hub	Output Load 1	DroidLink Master (ESP32 S3)	Vin / GND	12V Logic Power
	Secondary 12V Hub	Output Load 2	Syren 10A Dome Driver	Power Input (V+/GND)	12V Dome Motor Power
	Syren 10A Driver	Motor Out M1/M2	Greartisan Dome Motor	Motor Leads (+/-)	Modulated 12V Output
	5V Buck Converter	Output	Tertiary 5V Hub	Input	Inferred: not in list
	Tertiary 5V Hub	Output Load 1	Body Slave ESP32	5V / GND	Local 5V Logic Power
	Tertiary 5V Hub	Output Load 2	Pololu Maestro Mini 24	VCC / GND	Servo Rail / Board Power
	Tertiary 5V Hub	Output Load 3	LED Strips (Arms/Coins)	5V / GND	High-Density LED Power
	DroidLink Master	GPI04	Left Leg Flipsky ESC	Signal Input (w/ GND)	Left Throttle Command
	DroidLink Master	GPI05	Right Leg Flipsky ESC	Signal Input (w/ GND)	Right Throttle Command
	DroidLink Master	GPI021	Syren 10A Dome Driver	PWM / S1 Input	Dome Rotation Command
	DroidLink Master	GPI016	Drive RC Receiver	Signal, VCC, GND	Standard SBUS-A Input
	DroidLink Master	GPI017	Dome RC Receiver	Signal, VCC, GND	Standard SBUS-A Input
	DroidLink Master	GPI018	DFPlayer Mini MP3	RX, VCC, GND (x2)	Serial Sound Logic
	DFPlayer Mini	SPK_1 / SPK_2	Audio Speaker	Negative (-) / Positive (+)	Differential Audio Out
	Body Slave ESP32	GPI017	Pololu Maestro Mini 24	RX / GND	Local Servo Control Loop
	Body Slave ESP32	GPI04	LED Strips (Arms/Coins)	Data / GND	Dynamic LED Pixels Signal
	Pololu Maestro Mini	CH0 / CH1	Utility Arm Servos 1 & 2	Signal, VCC, GND	3-Wire Servo Interconnect
	Slip Ring Crossover	Dome Side Cap	Dome 5V/8A Buck Converter	Input (V+/GND)	18V Dome Feed
	Dome 5V/8A Buck	Output	2-in-5 Dome Lever Conn.	Input (V+/GND)	Stabilized Dome 5V Rail
	Dome 5V Hub	Output Load 1	AstroPixel Plus	Power Input (5V/GND)	Isolated Component Power
	Dome 5V Hub	Output Load 2	Pololu Maestro A	Power Input (5V/GND)	Isolated Component Power
	Dome 5V Hub	Output Load 3	Pololu Maestro B	Power Input (5V/GND)	Isolated Component Power
	Dome 5V Hub	Output Load 4	Magic Panel ESP32-C3	Power Input (5V/GND)	Isolated Component Power
	AstroPixel Plus	Serial 2 Pin T / G	Maestro A & Maestro B	RX & GND (Shared Bus)	Shared Serial Control Line